HajjSim Agent Studio: A Multi-Agent Swarm Intelligence Framework for Hajj Crowd Prediction and Adversarial Safety Analysis

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Abstract

The Hajj pilgrimage constitutes the most concentrated, time-constrained mass human movement globally, with over 2.5 million pilgrims navigating a rigid spatial-temporal ritual sequence. Traditional crowd management methodologies remain heavily reactive, frequently responding to critical density thresholds only after they materialize into physical bottlenecks. In this paper, we present HajjSim Agent Studio, a predictive digital twin platform utilizing Multi-Agent Swarm Intelligence. By modeling individual pilgrims as autonomous agents possessing a complex four-layer cognitive anatomy (Static Profile, Dynamic State, Memory, and Behavior Engine), the system simulates emergent crowd dynamics such as density waves and panic cascadesacross a GraphRAG spatial mapping of Mina and the Jamarat. Utilizing an event-driven, macro-temporal simulation engine where a single tick represents a discrete ritual stage rather than a fixed time interval, HajjSim allows operators to inject adversarial hazards and observe swarm resilience. The results demonstrate that predictive agent-based modeling can successfully identify invisible stress accumulation before physical crushing occurs, fundamentally shifting operational strategy from reactive damage control to proactive safety engineering.

1 Introduction & Motivation

Over 2.5 million pilgrims traverse spaces never originally designed for such extreme density during the five days of Hajj. The tragic 2015 Mina incident, which resulted in an estimated 2,400 casualties in a matter of minutes, underscores the catastrophic potential of minor spatial anomalies triggering massive, localized panic cascades.

Current crowd control methodologies rely heavily on closed-circuit observation, static mathematical capacities, and post-incident response. HajjSim Agent Studio addresses a fundamental computational challenge: the ability to proactively rehearse the complex Hajj sequence thousands of times in a digital environment before a single pilgrim arrives. Inspired by swarm intelligence paradigms, this project seeds a high-fidelity virtual world using baseline demographic constraints, spawns thousands of memory-bearing autonomous agents, and observes the emergent crowd behaviors that arise dynamically over the complex, multi-day sequence of rituals.

2 Related Work

To properly understand the architectural innovation of HajjSim, it is critical to examine how crowd movement has historically been simulated. In the study of pedestrian dynamics, researchers have primarily relied on mathematical and physics-based models, which treat humans as unthinking particles.

The foundational model in this space is the **Social Force Model (SFM)** (1). In the SFM, humans are subjected to simulated attractive and repulsive forces. They are “pulled” toward their destination and naturally “repel” away from walls and other people to maintain personal space. While this physics-based approach is excellent for simulating small-scale, urgent scenarios like evacuating a single room, it completely fails to capture the prolonged psychological complexity of the Hajj. Particles do not experience fatigue, they do not possess a multi-day schedule, and they do not share social memory.

To resolve these limitations, modern computational sociology utilizes **Agent-Based Modeling (ABM)** (2). In ABM, each entity is granted an independent decision-making loop. While ABM has been utilized in urban planning scenarios (3), HajjSim specifically adapts this framework for religious mass-gatherings. Table 1 highlights the differences between traditional models and our proposed architecture.

Simulation Model	Core Driver	Strengths	Limitations
Cellular Automata	Spatial grid constraints	Computationally efficient for very large macroscopic flows.	Too rigid for fluid, high-density crowd dynamics and individual pathing.
Social Force (SFM)	Physics equations (repulsion/attraction)	Excellent for modeling physical collisions and local personal space.	Lacks cognitive states; particles do not experience fatigue, panic, or memory.
HajjSim (ABM)	Cognitive *Perceive* → *Decide* → *Act* loops	High fidelity; tracks psychological stress, social memory, and ritual schedules.	Computationally heavy; requires macro-tick optimization for massive populations.

Table 1: Comparative analysis of historical crowd simulation models against the proposed HajjSim architecture.

3 Agent Architecture & Environment Design

3.1 The Four-Layer Pilgrim Anatomy

To ensure high-fidelity emergent behavior without hard-coding specific movements, each Pilgrim Agent is defined by a multi-dimensional data schema:

- **Static Profile:** Immutable traits assigned at initialization, including demographic data, nationality, and a base mobility index (e.g., accommodating elderly or wheelchair-bound pilgrims).
- **Dynamic State:** Real-time physiological variables, notably accumulated fatigue and psychological stress. Stress acts as a critical threshold metric, spiking non-linearly when local density exceeds 4 persons/m^2.
- **Memory Component:** Storage of immediately traversed paths to prevent routing loops, and social memory to track the location of group leaders, simulating herd behavior.
- **Behavior Engine:** The primary logic controller executing a continuous *Perceive* → *Decide* → *Act* loop. It balances macro-level ritual goals with micro-level environmental pressures.

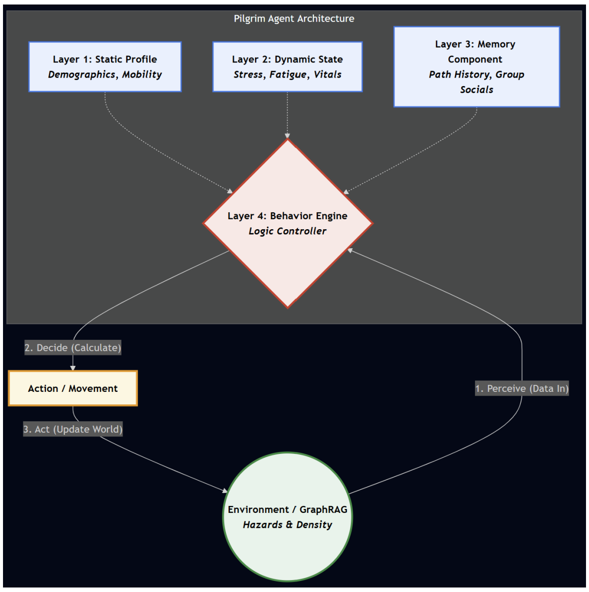

Figure 1: The 4-layer cognitive architecture of a Pilgrim Agent and the behavior loop.

3.2 Environment and Spatial Graph

The digital environment is not a passive canvas but an active data participant. Constructed using a GraphRAG approach, spatial nodes (such as specific Mina camps or the Jamarat pillars) emit atmospheric data. Furthermore, adversarial hazards such as stalled vehicles or dropped luggage can be dynamically injected into the simulation by the user to test the swarm's adaptive routing capabilities.

4 Methodology

HajjSim Agent Studio operates on a decoupled, highly modular architecture. The backend is built in Python utilizing JSON data stores to maintain state, while the frontend is an interactive Single-Page Application (SPA) utilizing JavaScript, HTML5, and CSS3 to render the operational dashboard.

4.1 Macro-Temporal Stage Engine

A major architectural divergence in HajjSim is its approach to simulated time. Unlike micro-simulations where a time "tick" represents mere seconds, HajjSim utilizes an event-driven scheduling methodology. In our system, **1 tick represents a distinct ritual window or stage**.

While some ticks encompass an entire day, high-density operational days are subdivided to accurately capture rapid crowd fluctuations. As detailed in Table 2, the complex movements of the 10th of Dhul-Hijjah are split across three separate ticks. This enables the system to process massive populations over the full sequence without computational overload, whilst maintaining strict logical fidelity to the pilgrim journey.

Tick	Ritual Stage	Day Label
0	Pre-start state	Arrival in Jeddah
1	Tawaf Al-Qudoum	Arrival in Jeddah
2	Stay in Mina	8th Dhul-Hijjah
3	Wuquf at Arafah	9th Dhul-Hijjah
4	Stay in Muzdalifah	Night of 9th to 10th
5	**Ramy at Aqaba**	**10th Dhul-Hijjah**
6	**Sacrifice**	**10th Dhul-Hijjah**
7	**Tawaf al-Ifadhah**	**10th Dhul-Hijjah**
8	Night in Mina	Night of 10th
9–11	Mina Stays	11th–13th Dhul-Hijjah
12	Tawaf al-Wadaa	Post-Hajj

Table 2: The macro-temporal ritual schedule mapped to simulation ticks.

5 Experimental Results & Discussion

Theoretical execution of the model demonstrates successful identification of emergent bottlenecks. By advancing the ritual-stage ticks, the simulation accurately predicts rapid stress accumulation during peak transition windows, most notably the massive migration following the Wuquf at Arafah (Tick 3) into Muzdalifah (Tick 4).

The integration of Leaflet.js heatmaps on the frontend dashboard provides immediate visual feedback to operators. The system successfully translates dense matrix calculations of individual agent stress and fatigue into an accessible, visual Severity Index. When an adversarial hazard is introduced at a chokepoint during Tick 5 (Ramy at Aqaba), the system successfully visualizes the resulting "shockwave" of crowd density, confirming the hypothesis that minor physical blockages yield exponential psychological stress across the swarm.

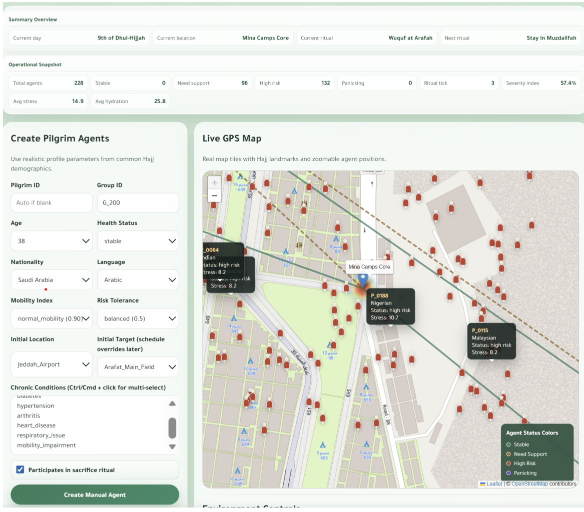

Figure 2: Live dashboard visualization showing agent clustering and heatmaps indicating high-stress zones.

6 Conclusion & Future Work

HajjSim Agent Studio successfully bridges the gap between theoretical swarm intelligence and applied crowd management logistics. By utilizing a modular agent architecture and an event-driven ritual-stage tick engine within an interactive web dashboard, it proves the viability of predictive safety testing for the Hajj. Future iterations will replace the in-memory JSON state with persistent relational database solutions, and integrate true live LLM endpoints to dynamically govern the adversarial behavioral overrides during simulated emergencies.

Known Project Limitations

The current system relies on an in-memory state repository, capping the maximum concurrent agent count before browser rendering degradation occurs. Furthermore, while the behavioral algorithms are logically structured, they presently rely on generalized theoretical thresholds for panic induction rather than live, empirically validated biometric feedback.

Authorship Statement

The development of HajjSim Agent Studio was a highly collaborative effort. **Joman Alsharbini** and **Sara Bin Shayhun** spearheaded the frontend dashboard design, including the Leaflet map integrations, visual heatmaps, and UI/UX flows. **Ahmad Baabod** developed the core API architecture, system integration, and state management. **Osama Abutaha** designed the theoretical 4-layer agent anatomy, the macro-tick simulation logic, and the backend data structures.

The team extends special thanks and gratitude to **Dr. Naila Marir** for her invaluable instruction, guidance, and support throughout the project's requirements and execution.

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